02 Assignment2 Design Contracts Schema

ShipTrack (shipment1212). TeamID: TODO | Name/Roll: TODO | Date: 2026-09-08.

Assignment 2 — Services, Contracts & Schema (mapped to ShipTrack)

TeamID: TODO | Name/Roll: TODO

CampusEats asked for design.pdf, services.drawio+png, schema.drawio+png+sql. Same content lives here:

1. Capabilities (≥6 verbs)

From docs/architecture.md:52-65 + README.md:40-53: register/login/RBAC, manage customer+carrier profiles, getQuote, createShipment, trackPublic/listHistory, recordEvent/latest, updateStatus/assignCarrier, currencyConvert/countryInfo, analytics aggregate, audit, notify, GraphQL dashboard.

2. Service design (3–6 boxes → we have 12, boundary holds)

Diagram: docs/architecture.md:5-48 (mermaid flowchart) + editable docs/diagrams/use-case-diagram.drawio + .png, data-flow-diagram.drawio. Each box owns its data, no sharing:

Service	Port	Owns (DB)
auth	8001	Postgres auth_db
user	8002	Postgres user_db
carrier	8003	Postgres carrier_db
shipment	8004	Postgres shipment_db
tracking	8005	Mongo tracking_db
pricing	8006	MySQL pricing_db
location	8007	MySQL location_db
crm	8008	Mongo crm_db
audit	8009	Mongo audit_db
notification	8010	— (SMTP/log)
graphql	8011	— (aggregator)
gateway	8000	— (routing)

Cross-service arrows: Shipment → Pricing: POST /pricing/quote, Shipment → Tracking: POST /tracking/:id/events (services/shipment-service/src/clients/), Pricing → Location: SOAP GetCountryInfo (services/pricing-service/src/soap/countryClient.js), GraphQL → Shipment/Tracking/User: REST. Member per service: TODO (add names to boxes before submitting).

3. Contracts (operation, input, output, errors — no tables/SQL)

In services/*/openapi.yaml (12 files). Two main examples:
- shipment-service (7 ops): POST /shipments in {origin, destination, weightKg, packageType, currency} out 201 Shipment, errors 400/401/403/409/503; GET /{id} out 200, errors 403/404; PATCH /{id}/status in {status, expectedVersion} errors 412/422.
- tracking-service (3 ops): POST /{trackingId}/events in {status, location, note} out 201 TrackingEvent, errors 400; GET /{trackingId}/events?cursor&limit out 200; GET /{trackingId}/latest out 200, errors 404.
Swagger: http://localhost:18000/api-docs + per-service http://localhost:<8001-8010>/api-docs.

4. Central operation — placeOrder → POST /api/v1/shipments createShipment

Code: services/shipment-service/src/controllers/shipmentController.js:11-18 + src/services/shipmentService.js. Spec in services/shipment-service/openapi.yaml (/api/v1/shipments POST).
- In: {origin{address,city,country}, destination{…}, weightKg, packageType, currency} + JWT + Idempotency-Key header.
- Out: 201 {id, trackingId, quotedPrice, currency, estimatedDeliveryDate, version}.
- Errors: 400 malformed, 401/403 auth, 409 same key+different body (common/src/idempotency.js:29), 503 pricing/tracking down.
- Hidden: Postgres table, UUID/trackingId gen, pricing SOAP call, tracking best-effort POST, optimistic-concurrency version.

5. Database schema (ER + CREATE TABLE)

- ER: docs/diagrams/er-diagram.drawio + .png, schema-diagram.drawio + .png, docs/er-diagrams.md.
- SQL: infra/postgres-init/01-create-databases.sql, infra/mysql-init/01-create-databases.sql, per-service src/db/migrate.js.

- Rule holds: no table belongs to two services (see ownership table above). Verify: docker exec -it shiptrack-node-postgres psql -U shiptrack -d shipment_db \dt (creds RUNBOOK.md §4).

6. Five service properties

Reachable (own port + gateway route) · self-contained (own DB, no cross store import) · contract (openapi.yaml + /api-docs) · independent (own Dockerfile) · loosely-coupled (only createServiceClient HTTP via env URLs in common/src/httpClient.js). Known fix: frontend nginx.conf proxy_pass must be repointed on Render (deployment guide).